Assignment-3 1923111 77

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Task 1: Content - free Grammar Design and Derivation

Content -free Grammar:- TO Enfore standard mathamatical precondence, we structure the grammer into expressions (E), Terms (T), Eacfor(F).

Terminals: { +, * , (,), a, b, c }

non-terminals:- { E, T, F }

Start Symbols: E

* production Rule:-

$E \rightarrow E+T/T$

$T \rightarrow T*F/F$

$F \rightarrow (E)/a/b/c$

LMD → a+b*c

$E \rightarrow E+T$

$\Rightarrow T+T$

$\rightarrow F+T$

$\Rightarrow a+T$

$\Rightarrow a+T*F$

$\Rightarrow a+F*F$

$\Rightarrow a+b*F$

$\Rightarrow a+b*c$

RMD → a+b*c

$E+T*F$

$E+T*c$

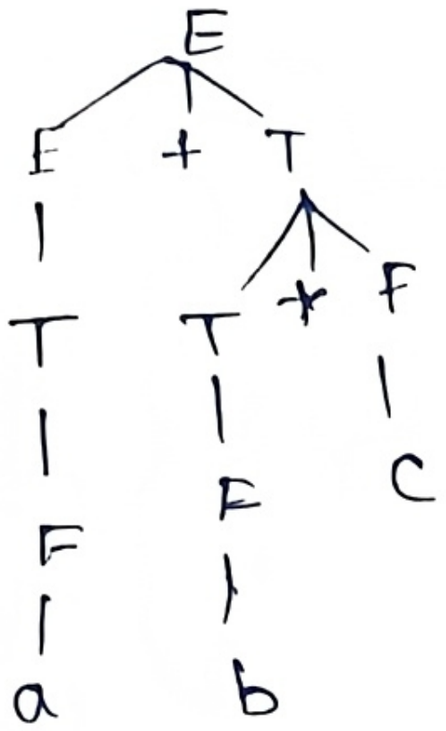
$E+F*c$

$E+b*c$

$T+b*c$

$a+b*c$

Parse Tree:



Task 2: Simplification of context Free Grammar original Grammar:

$S \rightarrow AB \mid a$

$A \rightarrow aA \mid \epsilon$

$B \rightarrow BB \mid c$

$C \rightarrow C$

1. Identify and eliminate nullable variable:-

* Null Variable: A (since $A \rightarrow \epsilon$)

* Remove $A \rightarrow \epsilon$ and add productions substituting A with ϵ where, if appears on right hand side

New Grammar:

$S \rightarrow AB \mid B \mid a$

$A \rightarrow aA \mid a$

$B \rightarrow bB \mid c$

$C \rightarrow C$

Identify and Eliminate unit productions:

Unit productions : $S \rightarrow B$ and $B \rightarrow c$

*Eliminate $B \rightarrow c$: Replace it with the right-hand side of c ($c \rightarrow c$). This updates B to $B \rightarrow bBc$

* Elimate $S \rightarrow B$: Replace it with the update of right-hand side of $B(bBc)$

* New Grammar:

$S \rightarrow AB / bB / c / a$

$A \rightarrow aA / a$

$B \rightarrow bB / c$

$C \rightarrow C$

3. Final simplified CFG:-

$S \rightarrow AB / bB / a / c$

$A \rightarrow aA / a$

$B \rightarrow bB / c$

TASK 3:- Conversion into chomsky normal form.

Original Grammar:-

$S \rightarrow aAB / B$

$A \rightarrow aA / a$

$B \rightarrow bB / b$

1. Identify non-CNF :- CNF requires all produ
   to be V → W or V → T. The production S → aAB. Orig
   A → aA, and B → bB.
2. Introduce variable for terminal symbol:-
   S → XaAB | b
   A → XaA | a
   B → XbB | b
3. Break productions containing more than variable.
   the production S → XaAB has three variables.
   Y → AB to split.
   S → XaY
   Y ⇒ AB
4. Final chomsky normal form Grammar.
   S → XaY | b
   Y → AB
   A → XaA | a
   B → XbB | a.
   Xa → a
   Xb → b

Task 4: Conversion into Greibach normal form:

Original Grammar:

$S \rightarrow AB/a$

$A \rightarrow aA/a$

$B \rightarrow bB/a$

1. Identify non GNF productions:-

* $A \rightarrow aA/a$ and $B \rightarrow bB/b$ already satisfy GNF

* $S \rightarrow AB$ violates GNF because it starts with a non-terminal (A).

2. Final Greibach Normal Grammar:

$S \rightarrow aAB / aB/a$

$A \rightarrow aA / a$

$B \rightarrow bB / b$

3. Derivation of string:

$S \Rightarrow aAB$

$\Rightarrow aaB$

$\Rightarrow aab.$

Task 5:- content-free language, properties and pumping lemma.

$$L = \{a^n b^n c^n \mid n \geq 1 2\}$$

1. Determine, weather the language is content free:-
The given language is not content free because if requirs equal no. of a's, b's and c's.

2. state the pumping lemma for CFC

$S = uvwxy$

1. $|vwx| \leq p$
2. $|vx| \geq 1$
3. $uv^i wx^i y \in L$ for $i \geq 0$

$S = a^p b^p c^p$

clearly $S \in L$

for $i=0$ (or) $i=2$

$uv^i wx^i y \notin L$

The pumping lemma conditions is violated because the pumped string does not contains

$L = \{ a^n b^n c^n \mid n \geq 1 \}$ is not CFL.